



Wine Quality Machine Learning Report

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Introduction

This report aims to predict wine quality and identify the key features influencing it using the UCI Wine Quality dataset. Wine quality is important for ensuring product consistency and customer satisfaction. To achieve this, we applied three approaches including Regression, Classification and Unsupervised Learning. Regression models like Linear Regression and Random Forest are used to predict the continuous quality score. Classifications such as Logistic Regression and Random Forest Classifier group wines into quality categories. Unsupervised Learning techniques like Principal Component Analysis (PCA) and K-means identify patterns and group similar wines.

Data Preprocessing

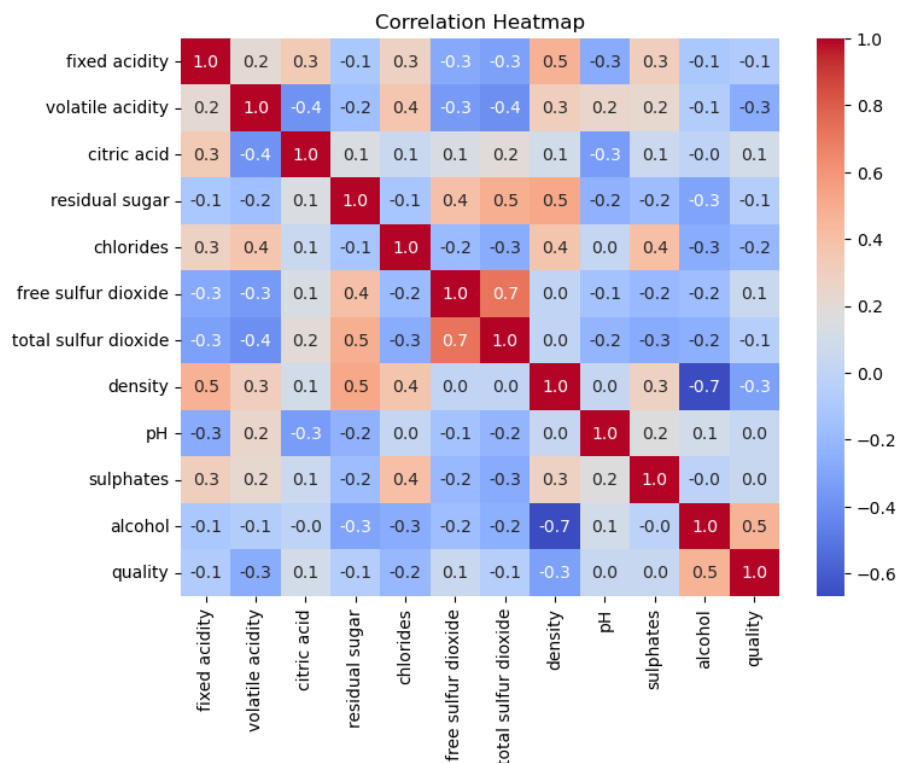
First, we need to combine the red and white wine datasets into one dataframe for predicting both wine quality and create a new column called 'type' to distinguish the wine type. To avoid inaccurate and biased results, checking missing values is essential and we note that there is no missing value in this dataset. Then, duplicates may lead to data redundancy so a total of 1,177 duplicate rows is removed and reducing the dataset size to 5,320 observations. Next, categorical variable 'type' is encoded into binary digits 0 (red) and 1 (white) to keep the data efficient thereby improving predictive accuracy. Before splitting the data frame, we should divide the data into feature (X) and target variable(Y). The dataset is then split into 80% training data and 20% testing data to ensure sufficient data for model training and maintaining reliable evaluation. Last, StandardScaler is fitted on the training data and applied to both the training and testing sets.

Exploratory Data Analysis (EDA)

EDA helps us to understand patterns, trends and relationships of dataset through statistical tools and visualization.

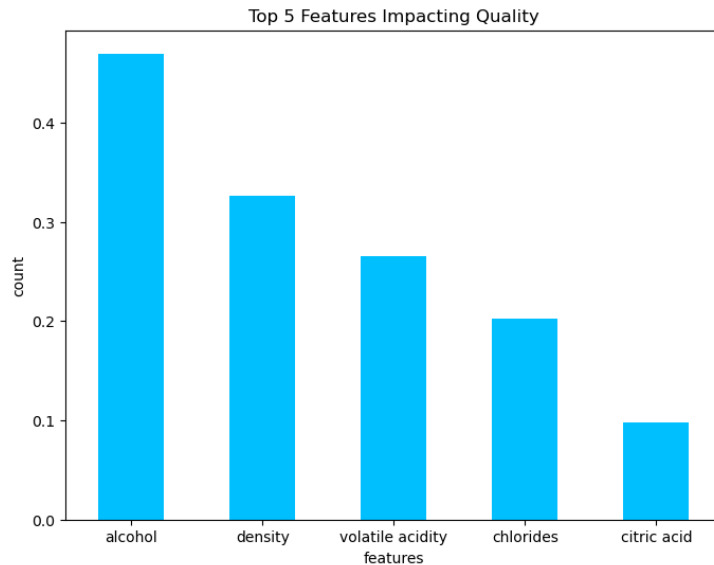
1. Correlation heatmap

This correlation heatmap shows that most attributes have weak to moderate relationships with wine quality. Alcohol is positively correlated with quality while volatile acidity and density show negative correlations. Strong correlations between some features may cause potential multicollinearity. However, model performance decreased in both Regression models after related features were dropped. This demonstrates that they are still useful for prediction and removing them does not always improve model performance.



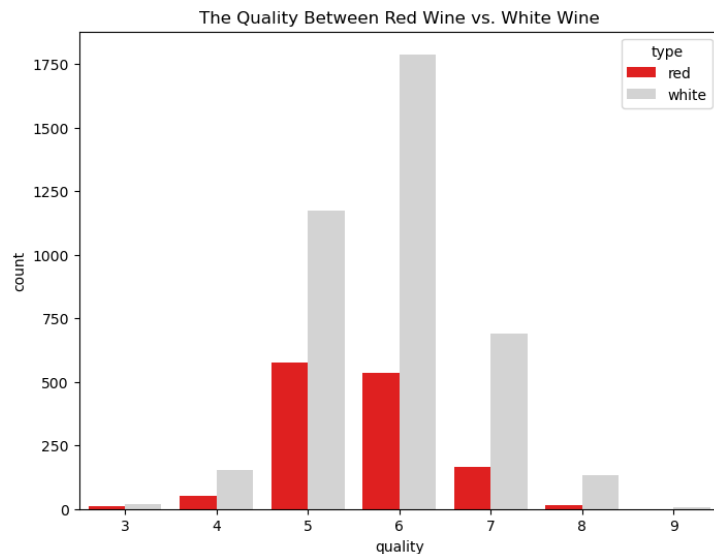
2. Top 5 Features Impacting Quality

This chart indicates the top five features that have the strongest correlation with wine quality in the dataset. Among these attributes, alcohol has the highest relationship with quality which affects the chemical balance and taste of wine. However, these results may differ from model-based feature importance which will be discussed in the Classification section.



3. Quality Between Red Wine vs. White Wine

This is a quality scores distribution graph for red and white wines. Most wines are concentrated around quality scores of 5, 6 and 7 (average quality). This dataset contains more white wine samples as white wine appears more frequently than red wine in most quality categories. We note that there is a serious data imbalance while high-quality wines in scores 8 and 9 are relatively rare.



Regression

Regression is a supervised learning technique used to predict continuous numerical values by learning the relationship between features (X) and target (Y). In this project, we applied Linear Regression and Random Forest Regression. Their performance is compared using R^2 , RMSE and MAE.

Linear Regression

Linear regression is a simple model that only assumes a linear relationship. Since we have two or more independent variables, multiple linear regression is suitable for this dataset.

Linear Regression R^2 :	0.2840061033445316
Linear Regression RMSE:	0.7447098132295159
Linear Regression MAE:	0.573648696090272

Interpretation:

The R^2 value of 0.2840 means that the model can explain 28.4% of the variation in wine quality which is relatively low and the relationship between X and Y is not purely linear. The RMSE value of 0.7447 shows that the prediction error is around 0.75 which is considerably high as the wine quality is measured in small integer values. The MAE value of 0.5740 indicates that the average absolute error between predicted and actual values is about 0.57 which predict accuracy moderately.

Conclusion:

Although Linear Regression model cannot explain much of the variation in wine quality and it cannot capture complex relationships in the data, it is still included as a baseline model which provides easy interpretation to compare with advanced models. With baseline, we can easily observe the better performance of a complex model. Additionally, its low R^2 shows that the relationship is not linear which justifies using advanced models like Random Forest Regression.

Random Forest Regression

Random Forest is an ensemble learning model that combines multiple decision trees to produce more accurate and stable predictions. It can be used effectively with dataset that contains many input features and complex relationship like nonlinear relationship between features and the target variable.

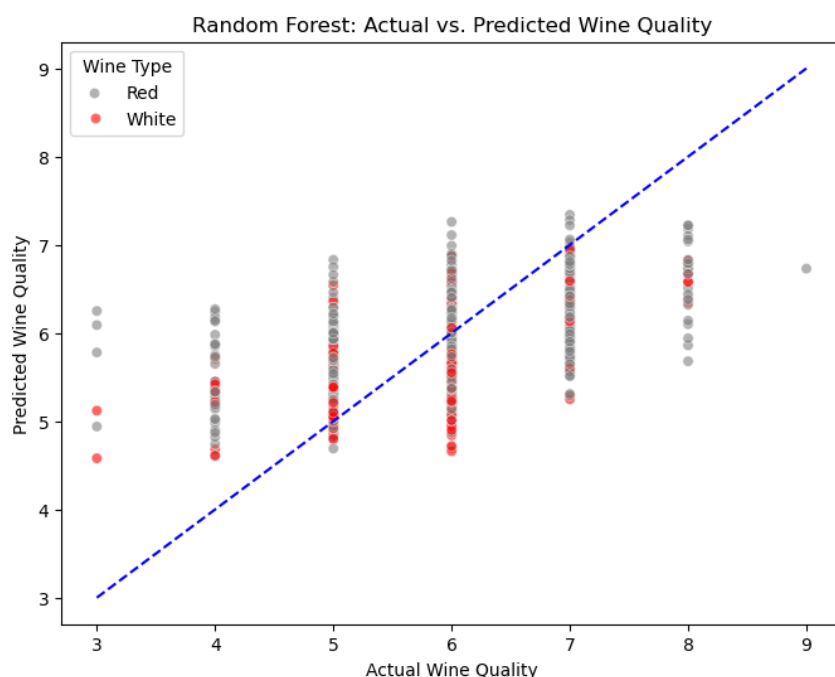
Random Forest Regression R^2 :	0.37008692205233007
Random Forest Regression RMSE:	0.6985101626808453
Random Forest Regression MAE:	0.5330357142857143

Interpretation:

The R^2 value of 0.3700 means the model explains about 37% of the variation which is higher than linear regression even though it is still a relatively low variation. This expresses that Random Forest fits the data better. The RMSE value decreases from 0.7447 to 0.6985, indicating that the prediction error is smaller. Similarly, Random Forest shows more accurate predictions since the MAE decreases to 0.5330.

These improvements justify that the relationship between features and quality is non-linear thus the complex interaction between variables cannot be captured by Linear Regression.

Graph:



This scatter plot shows that the model can make reasonably accurate predictions since most points are located near the dashed line. Points on the line would be perfect prediction, points above the line demonstrate model overpredicted and points below the line indicate model underpredicted. This plot proposes the nonlinear pattern, poor prediction for extreme values, severe underprediction of high-quality wines and both types of wines cannot be distinguished clearly.

Conclusion:

Random Forest performs better because it can observe complex patterns. However, the R^2 is still not high and one serious issue is the imbalance in the dataset. In the plot we can see that most wines fall into medium quality while very high and very low quality are rare. Because of this, the model learns more about the majority class and it is difficult to accurately predict extreme values.

Regression Conclusion:

To sum up, wine quality is difficult to predict precisely using only given chemical characteristics. While Random Forest improves performance by capturing non-linear relationships, the relatively low R^2 indicates that important factors are missing and the dataset is imbalanced. This means wine quality prediction cannot be fully explained by the current data.

Classification

Classification is a supervised machine learning technique where the model tends to predict the correct categories (Y) based on input data (X). In this project, Logistic Regression and Random Forest Classifier are applied. We can compare their performance through cross-validation accuracy, classification report and confusion matrix plot. In this section, wine quality is converted into three categories: Low (≤ 5), Medium (6), High (≥ 7)

Logistic Regression

Logistic Regression assumes linear relationship between the features and the log-odds of the dependent variable and estimates probabilities using sigmoid function. Since there are three categories, Multinomial Logistic Regression is applied.

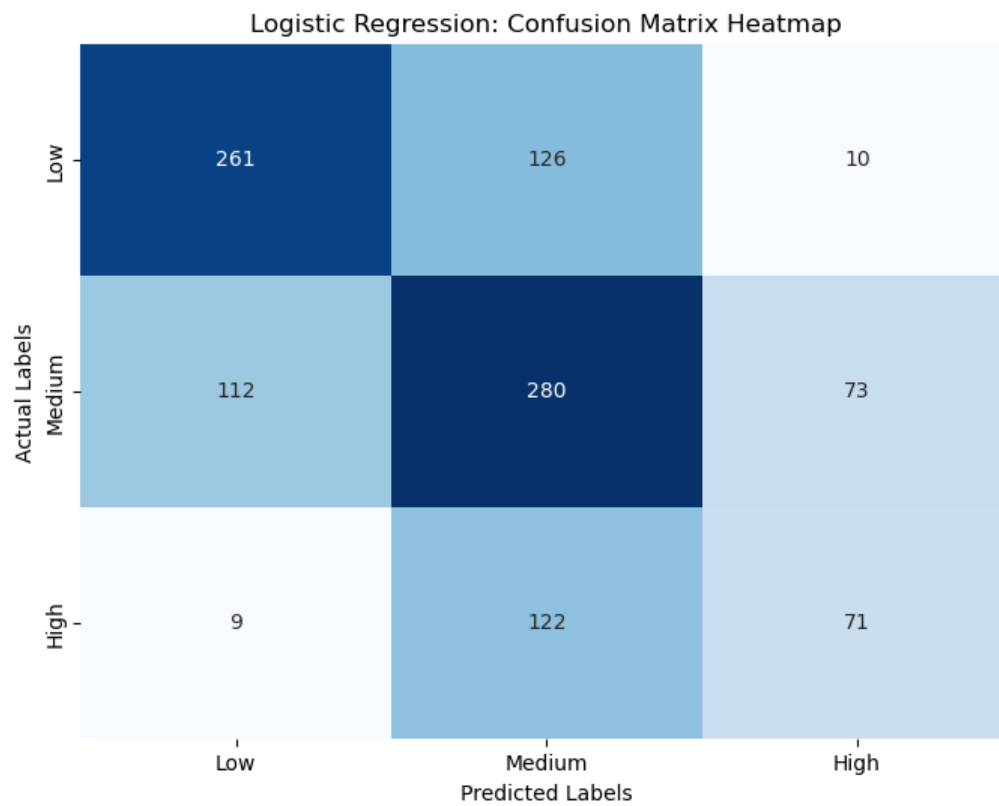
Accuracy: 57.52%

	Precision	Recall	F1-score	Support
Low	0.68	0.66	0.67	397
Medium	0.53	0.60	0.56	465
High	0.46	0.35	0.40	202
Accuracy			0.58	1064
Macro avg	0.56	0.54	0.54	1064
Weighted avg	0.57	0.58	0.57	1064

Interpretation:

The overall accuracy of 57.52% shows that the model has relatively low performance. The model predicts the low-quality wines reasonably well but still misclassifies some cases. Medium-quality wines are predicted with moderate performance, meaning many are correctly identified but errors still occur. High-quality wines are predicted poorly and the model struggles to detect rare top-quality wines. This indicates a bias toward more common classes and difficulty handling class imbalance.

Graph:



This plot proves the model performs best at predicting the medium class with 280 correct classifications. However, it struggles with class separation as many low (126) and medium (112) cases are misclassified. For High wines, most are incorrectly predicted as Medium (122) with only a few correctly identified (71). Overall, class imbalance and poor class separation cause biased predictions toward the majority class.

Conclusion:

Logistic Regression is included as a baseline model due to its simplicity and interpretability. It is not suitable for handling imbalanced and complex data especially when predicting rare classes.

Random Forest Classifier

Random Forest Classifier is an ensemble model that builds multiple decision trees and combines their prediction to improve accuracy and reduce overfitting.

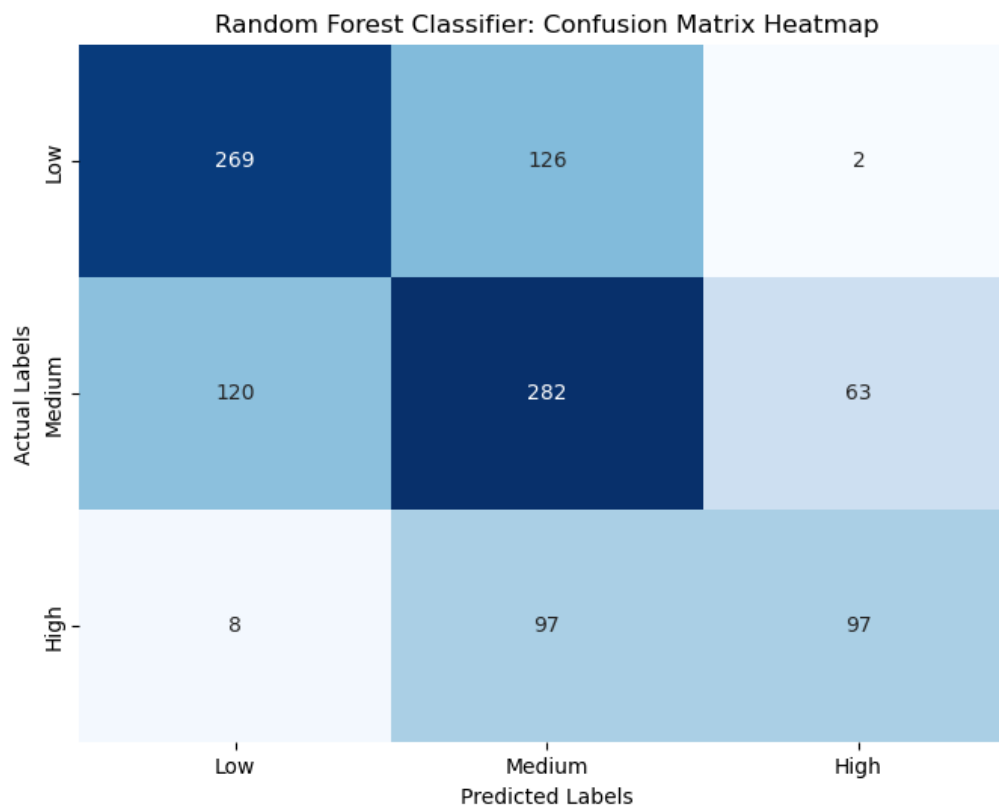
Accuracy: 60.90%

	Precision	Recall	F1-score	Support
Low	0.68	0.68	0.68	397
Medium	0.56	0.61	0.58	465
High	0.60	0.48	0.53	202
Accuracy			0.61	1064
Macro avg	0.61	0.59	0.60	1064
Weighted avg	0.61	0.61	0.61	1064

Interpretation:

The model achieves a moderate overall accuracy of 60.90% which shows better performance than logistic regression. It remains the best prediction of low-quality wines and improves detection of high-quality wines (F1-score=0.53) but misclassifications still occur. Medium quality wines are often confused with other classes, reducing prediction reliability. Random Forest handles non-linear relationships and class imbalance better by capturing complex patterns in the data.

Graph:

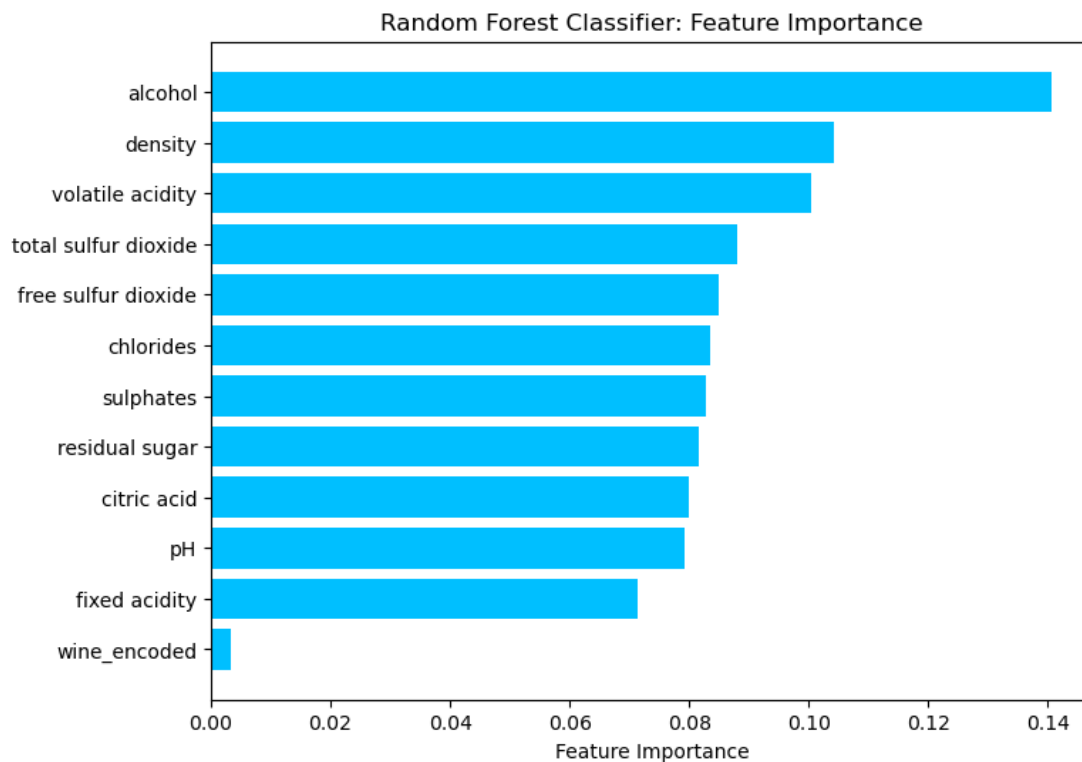


The model improves overall classification compared to Logistic Regression. It correctly predicts many low (269) and medium (282) wines though some misclassification still occurs between these classes. Notably, the correct prediction of high-quality wines increases from 71 to 97 instances. However, the model still suffers from class imbalance and confusion, particularly in misclassifying many high-quality wines as medium.

Conclusion:

Random Forest enhances prediction of wine quality, particularly for high-quality wines. Nevertheless, medium-quality wines remain frequently misclassified due to class imbalance. In addition, we tried to solve class imbalance by using class weighting but it did not lead to any significant improvement in the model results.

Graph:



This graph confirms that alcohol is the most influential predictor of wine quality same as the EDA results. Density and Volatile Acidity also rank highly which have similar patterns to correlation analysis. There are additional variables like total and free sulfur dioxide that are not involved in EDA plot, showing that the random forest models capture more complex relationships than simple EDA correlations.

Classification Conclusion:

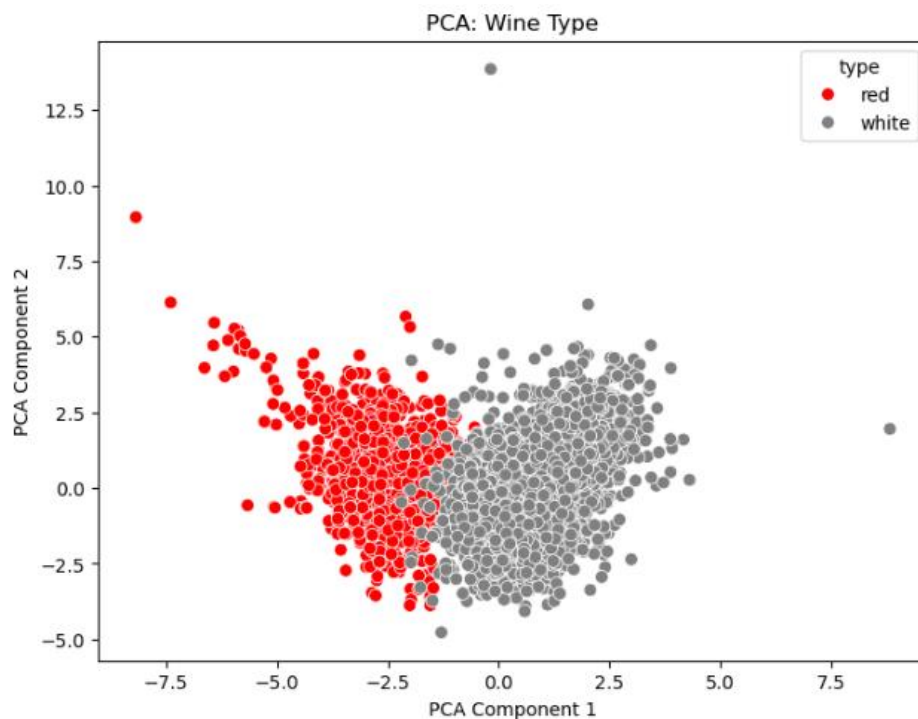
Overall, Random Forest performs better than Logistic Regression in predicting wine quality. It improves the detection of high-quality wines and captures more complex relationships between features, although class imbalance still affects the prediction performance.

Unsupervised Learning

Unsupervised learning works without labelled data and learns patterns itself by grouping similar data points or finding hidden structures in the dataset. Principal Component Analysis (PCA) and K-Means Clustering are the two popular techniques that can be applied in this project. Before modelling, we need to ensure all features are standardized and on the same scale.

Principal Component Analysis (PCA)

PCA reduces the number of features to two main components for easier visualization.



From the PCA plot, we can observe there is a clear separation between the red and white wines. In other words, one region is grouped as most red wines while white wines are in another. This indicates that the chemical features of both wines are significantly different. However, there is some overlap between two groups which means some wines have similar characteristics.

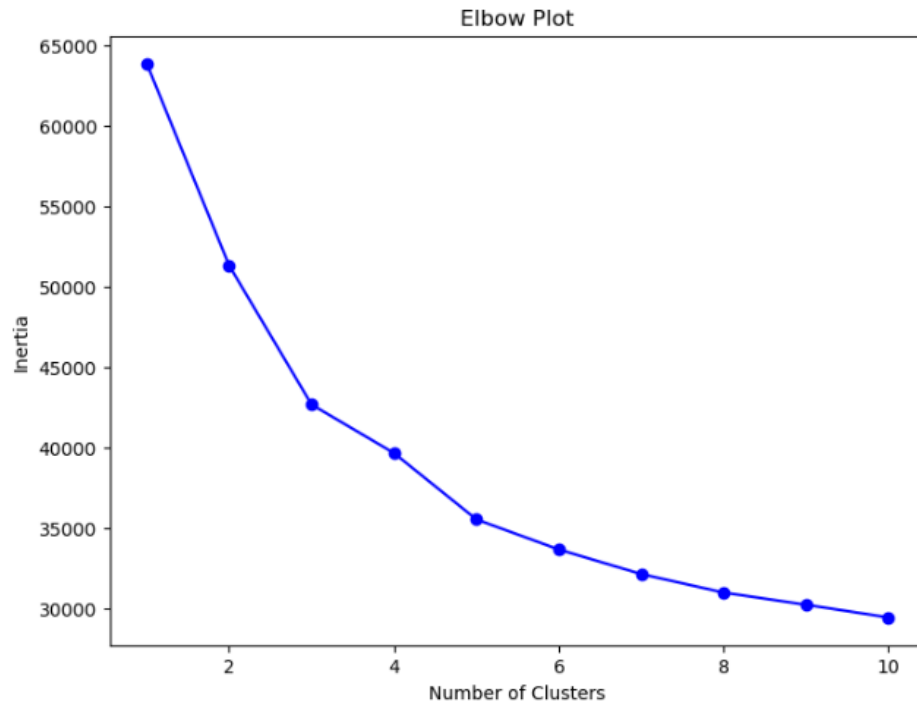
Conclusion:

Although PCA demonstrates that wine type is an important factor and the dataset has a clear structure, some overlap exists and the distinction is not perfect.

K-Means Clustering

This algorithm categorizes the items into 'k' clusters based on similarity.

First, we need to determine the number of clusters by utilizing the elbow method and silhouette score.



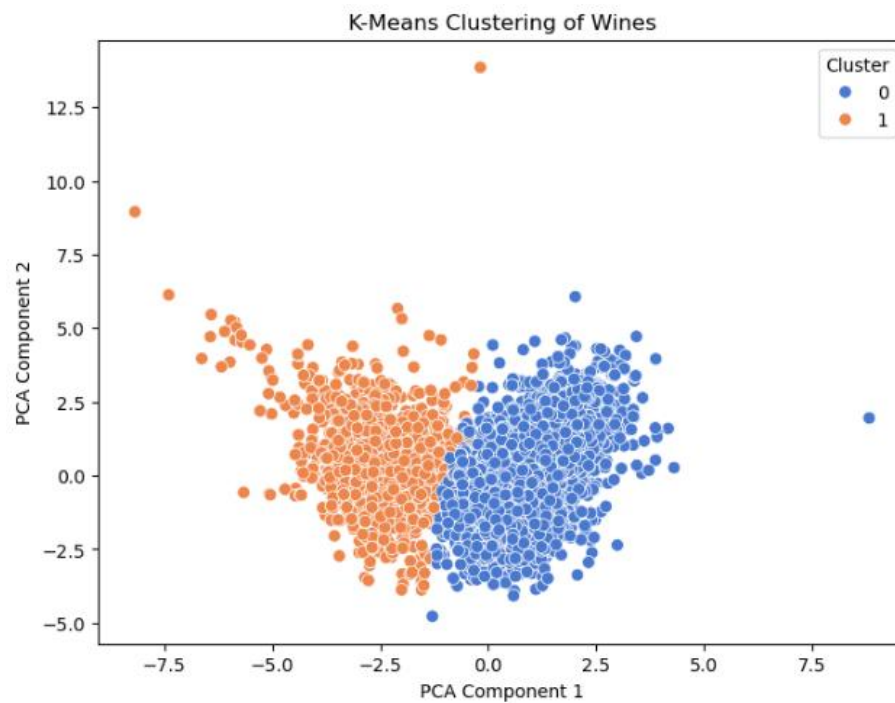
Silhouette score:

Number of clusters	Silhouette score (-1 to +1)
2	0.27
3	0.23
4	0.15
5	0.18

Interpretation:

The elbow plot shows the curve starts to flatten at $k=2$ which means the optimal cluster number is 2. In addition, the silhouette highest score is at $k=2$. This confirms that 2 clusters are the optimal choice.

Graph:



Cluster	Counts
0	3919
1	1401

Due to the optimal number of clusters ($k=2$), the points are divided into two groups (cluster 1 and cluster 2) clearly. This means K- Means successfully identified two main patterns in the data. Furthermore, we can observe that cluster 1 contains more points than the other, indicating that the dataset is not balanced. Moreover, the grouping is not perfectly distinct because some points are close between clusters.

Conclusion:

Based on the PCA results, the two clusters may correspond to red and white wines. This result shows that clustering is driven more by wine type than quality. However, the clusters do not clearly separate wine quality, indicating that the model cannot fully capture the variables that affect wine quality.

Unsupervised Learning Conclusion:

In short, PCA and K-Means show clear separation mainly by wine type, not quality. Wine quality is not well clustered as it is complex, overlapping and nonlinear feature relationship.

Discussion

Different models produce varied wine quality predictions depending on their ability to handle nonlinear relationship and significant class imbalance issues in the dataset. Random Forest achieves the best performance in both regression and classification because it can capture interactions between chemical features more effectively. Linear and Logistic Regression perform weaker because simple linear assumptions are not suitable for this dataset and they are purely for comparison as a baseline model. In Unsupervised Learning, PCA and K-Means reveal a clear separation by wine type rather than quality while quality differences are less distinct and harder for models to identify accurately.

Conclusion

In conclusion, this project aims to predict wine quality using multiple machine learning approaches and identify key influencing features. The results show that Random Forest models perform best by capturing nonlinear relationships between features while linear models are utilized as useful baselines for comparison. However, overall performance remains moderate due to class imbalance and limited feature information. In other words, wine quality is difficult to predict precisely using chemical attributes alone and a biased dataset. Unsupervised learning reveals that wine type is more distinguishable than quality as clustering is primarily driven by differences between red and white wines. To improve performance, we should include more extreme data (low and high quality wine data) and incorporate additional relevant features such as grape variety, vineyard location, vintage year, fermentation time, aging conditions and weather conditions to enhance prediction accuracy.

Reference

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4. [How to Interpret the Classification Report in sklearn \(With Example\)](#)
5. [What is Unsupervised Learning - GeeksforGeeks](#)
6. [What is Silhouette Score - GeeksforGeeks](#)